Signals and Systems Lecture-5: Unit Impulse and Comb function

Chadi Abou-Rjeily

Department of Electrical and Computer Engineering
Lebanese American University
chadi.abourjeily@lau.edu.lb

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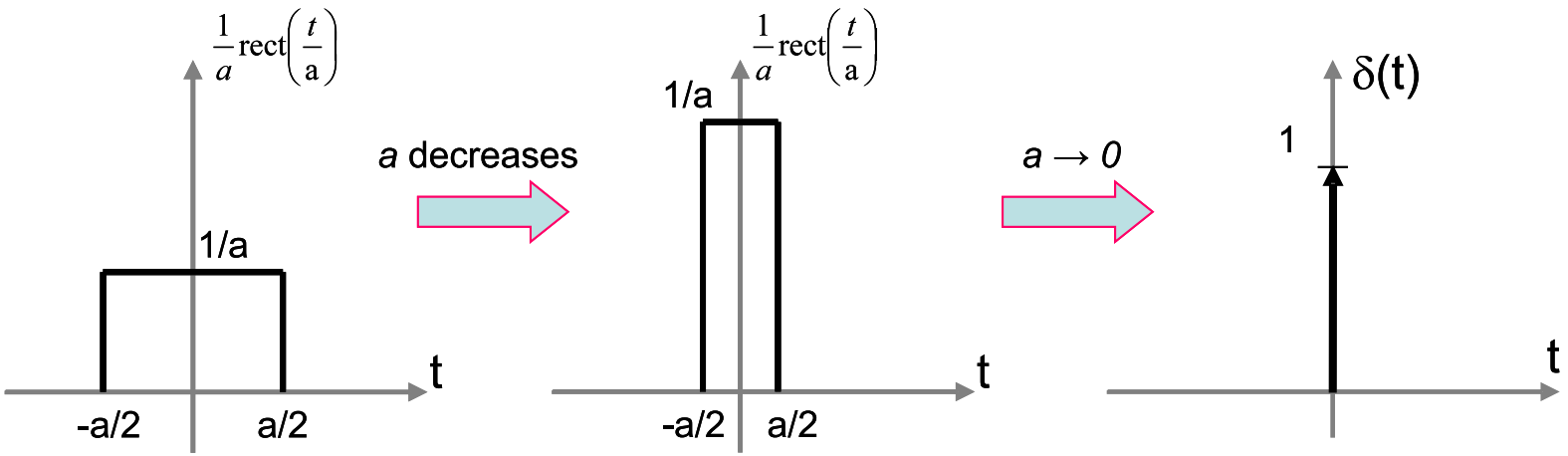
Unit Impulse (1)



- The unit impulse is defined as:

$$\delta(t) = \lim_{a \to 0} \frac{1}{a}\mathrm{rect}\left(\frac{t}{a}\right)$$

- Note that $\delta(t)$ is infinitely narrow and infinitely tall yet it integrates to 1:

$$\int_{-\infty}^{+\infty} \delta(t)\mathrm{d}t = 1$$

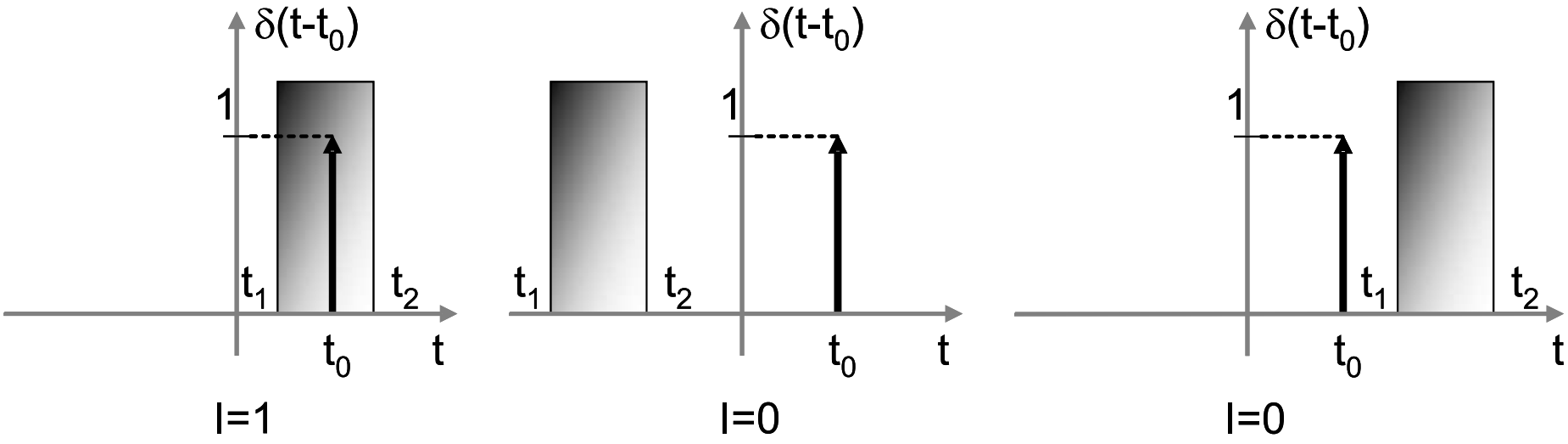
- In the graphical representation, the area under the impulse $\delta(t)$ (its weight) is indicated.

Unit Impulse (2)

Integration of the unit impulse:

$$\int_{t_1}^{t_2} \delta(t - t_0)\mathrm{d}t = \begin{cases} 1, & \text{for} \quad t_1 \leq t_0 \leq t_2 \\ 0, & \text{otherwise} \end{cases}$$

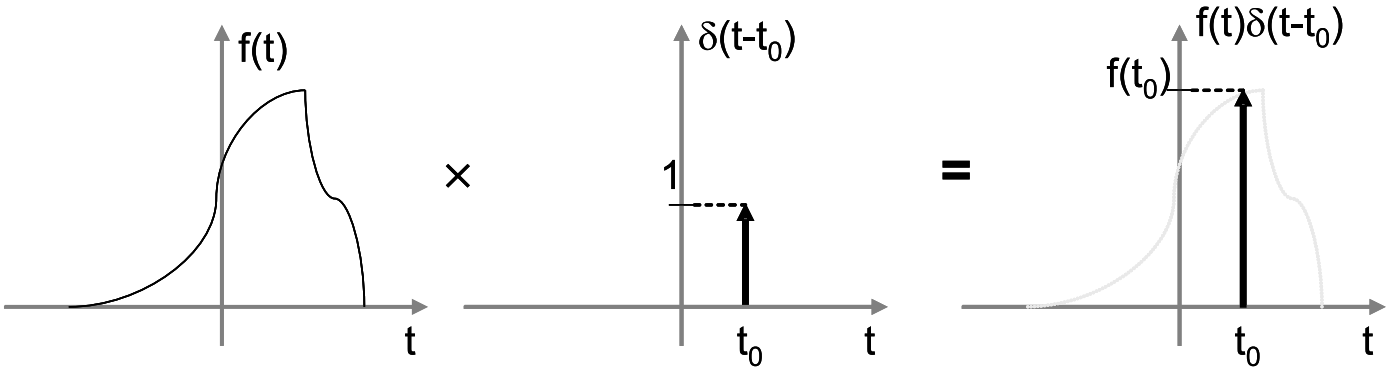
- The above property can be verified from the following figure where $I = \int_{t_1}^{t_2} \delta(t - t_0)\mathrm{d}t$:



Unit Impulse (3)

Multiplication by the unit impulse:

$$f(t)\delta(t-t_0) = f(t_0)\delta(t-t_0)$$



Unit Impulse (4)

- **Proposition**: The unit impulse verifies the following relation:

$$\int_{-\infty}^{+\infty} f(t)\delta(t-t_0)\mathrm{d}t = f(t_0)$$

- Proof:

$$\begin{aligned}\int_{-\infty}^{+\infty} f(t)\delta(t-t_0)\mathrm{d}t &= \int_{-\infty}^{+\infty} f(t_0)\delta(t-t_0)\mathrm{d}t \\ &= f(t_0)\int_{-\infty}^{+\infty} \delta(t-t_0)\mathrm{d}t \\ &= f(t_0)\end{aligned}$$

  - The first equality follows from the multiplication property of the unit impulse.
  - The third equality follows since the area under $\delta(t-t_0)$ is equal to 1.

Unit Impulse (5)

Proposition: the time-scaling of the unit impulse is defined by:

$$\delta(a(t-t_0)) = \frac{1}{|a|}\delta(t-t_0)$$

Proof:

- The unit impulse can be written as:

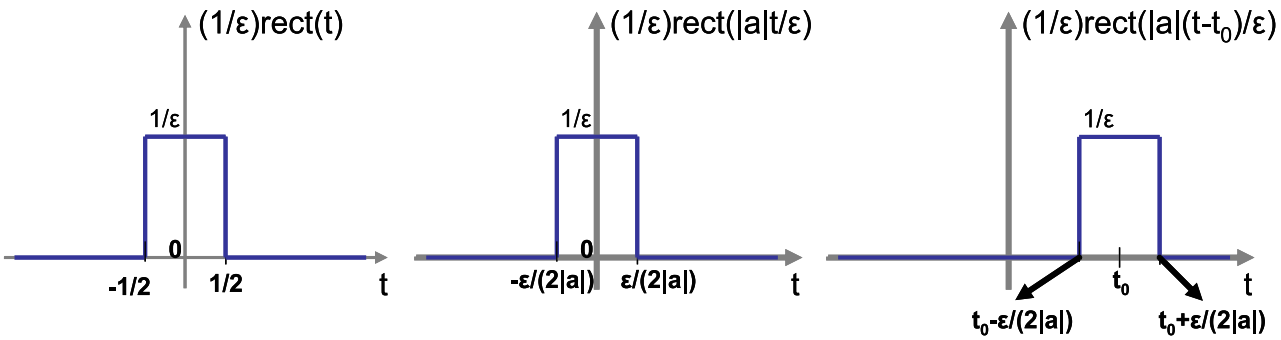
$$\delta(t) = \lim_{\epsilon \to 0} \frac{1}{\epsilon}\text{rect}\left(\frac{t}{\epsilon}\right)$$

implying that:

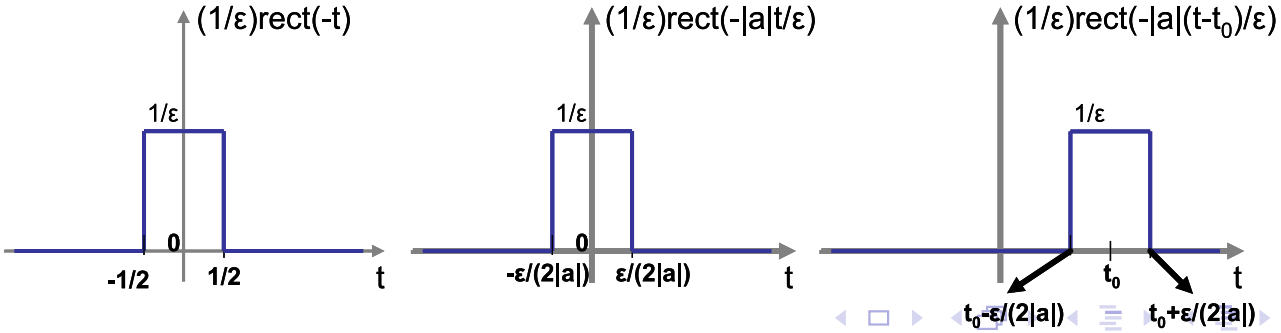
$$\begin{aligned}\delta(a(t-t_0)) &= \lim_{\epsilon \to 0} \frac{1}{\epsilon}\text{rect}\left(\frac{a(t-t_0)}{\epsilon}\right) \\ &= \begin{cases} \lim_{\epsilon \to 0} \frac{1}{\epsilon}\text{rect}\left(\frac{|a|(t-t_0)}{\epsilon}\right), & a > 0; \\ \lim_{\epsilon \to 0} \frac{1}{\epsilon}\text{rect}\left(\frac{-|a|(t-t_0)}{\epsilon}\right), & a < 0. \end{cases}\end{aligned}$$

Unit Impulse (6)

- For $a > 0$:
  - $t \to \frac{|a|}{\epsilon}t$: $\frac{1}{\epsilon}\text{rect}(t) \to \frac{1}{\epsilon}\text{rect}\left(\frac{|a|}{\epsilon}t\right)$.
  - $t \to t - t_0$: $\frac{1}{\epsilon}\text{rect}\left(\frac{|a|}{\epsilon}t\right) \to \frac{1}{\epsilon}\text{rect}\left(\frac{|a|}{\epsilon}(t - t_0)\right)$.



- For $a < 0$:
  - $t \to -t$: $\frac{1}{\epsilon}\text{rect}(t) \to \frac{1}{\epsilon}\text{rect}(-t)$.
  - $t \to \frac{|a|}{\epsilon}t$: $\frac{1}{\epsilon}\text{rect}(-t) \to \frac{1}{\epsilon}\text{rect}\left(\frac{-|a|}{\epsilon}t\right)$.
  - $t \to t - t_0$: $\frac{1}{\epsilon}\text{rect}\left(\frac{-|a|}{\epsilon}t\right) \to \frac{1}{\epsilon}\text{rect}\left(\frac{-|a|}{\epsilon}(t - t_0)\right)$.

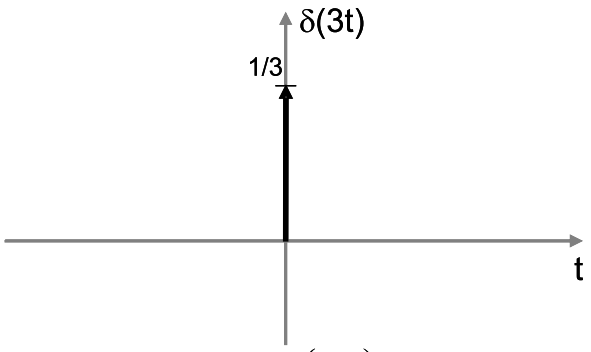


Unit Impulse (7)

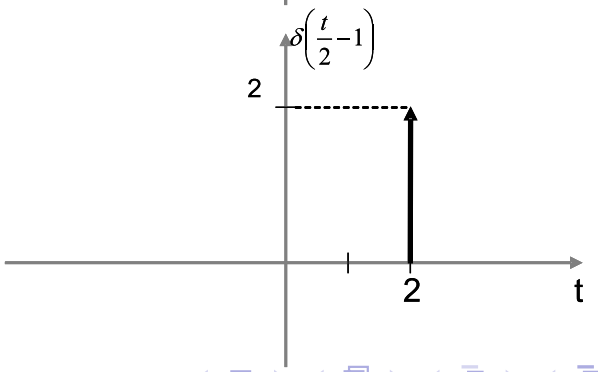
- In both cases, the function $\frac{1}{\epsilon}\text{rect}\left(\frac{a(t-t_0)}{\epsilon}\right)$ is centered at t_0 and has an area of $1/|a|$. Consequently, as $\epsilon \to 0$:

$$\delta(a(t-t_0)) = \lim_{\epsilon \to 0} \frac{1}{\epsilon}\text{rect}\left(\frac{a(t-t_0)}{\epsilon}\right) = \frac{1}{|a|}\delta(t-t_0)$$

- Examples:
  - $\delta(3t) = \frac{1}{3}\delta(t)$



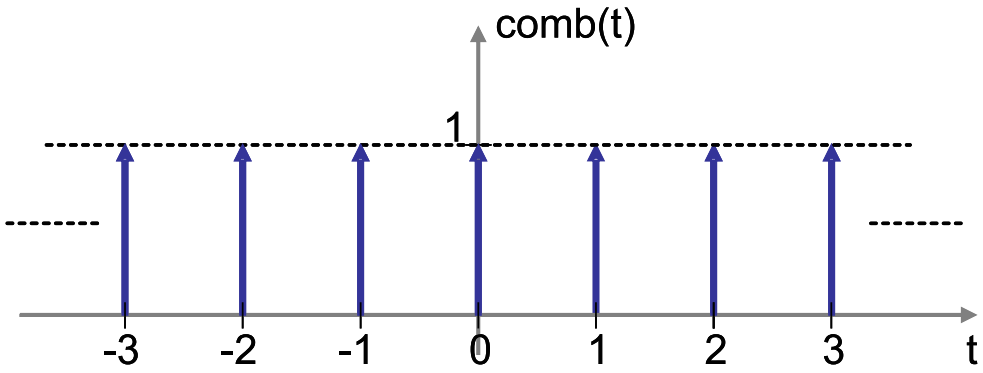
  - $\delta(\frac{t}{2}-1) = \delta(\frac{t-2}{2}) = \delta(\frac{1}{2}(t-2)) = 2\delta(t-2)$



The Comb function (1)

- The comb function is related to the unit impulse by:

$$comb(t) = \sum_{n \in \mathbb{Z}} \delta(t - n)$$



The Comb function (2)

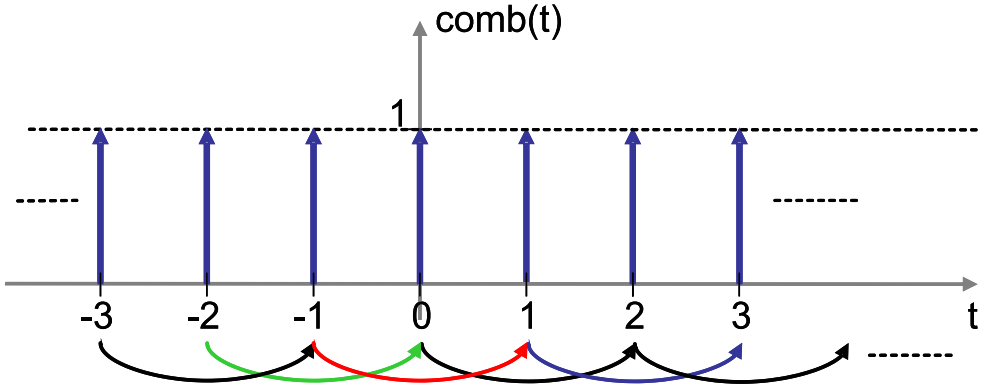
- $comb(t)$ admits the y-axis as an axis of symmetry ($comb(t)$ is an even function):

$$comb(-t) = comb(t)$$

- The comb function is periodic with period 1. Consequently, it is invariant under translations by integer values:

$$comb(t - m) = comb(t) \;\; \forall \;\; m \in \mathbb{Z}$$

  - For example, for $m = 2$:

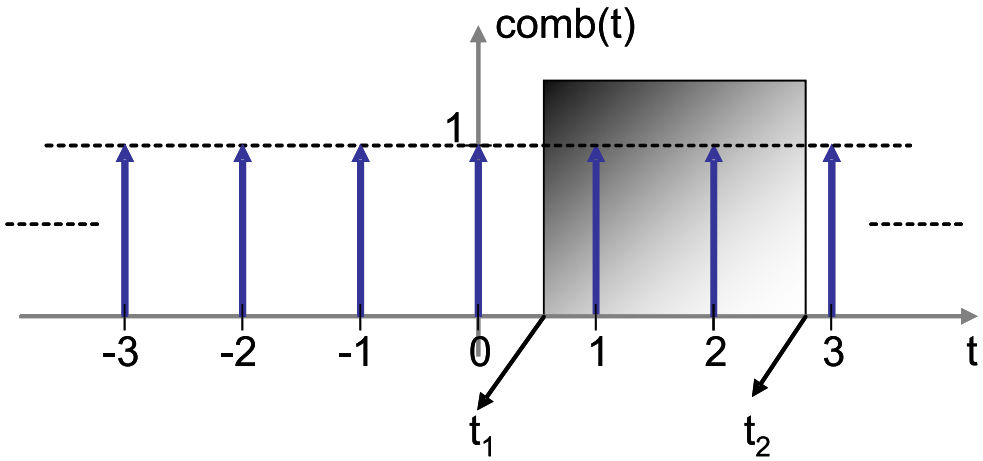


The Comb function (3)

- The comb function verifies the following relation:

$$\int_{t_1}^{t_2} comb(t)\mathrm{d}t = n$$

where n is the number of integers in the interval $[t_1 \;\; t_2]$.



- Example:

$$\int_{n-\frac{1}{2}}^{n+\frac{1}{2}} comb(t)\mathrm{d}t = 1 \;\; \forall \;\; n \in \mathbb{Z}$$

There is only one unit impulse in the boundaries of the above integral. This impulse is $\delta(t-n)$.

The Comb function (4)

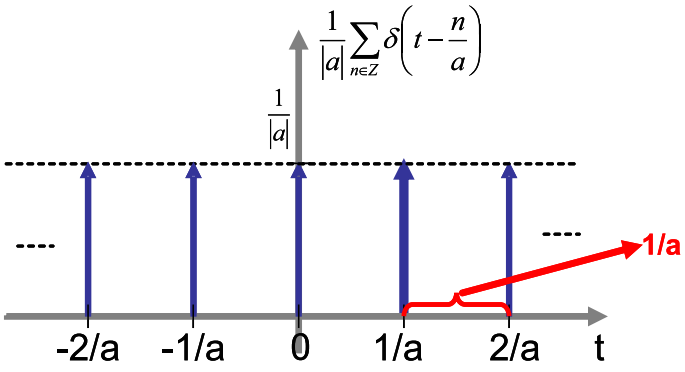
- **Proposition**: $comb(at)$ is given by:

$$comb(at) = \frac{1}{|a|}\sum_{n\in\mathbb{Z}} \delta\left(t - \frac{n}{a}\right)$$

- *Proof:*

$$comb(at) = \sum_{n\in\mathbb{Z}} \delta(at - n) = \sum_{n\in\mathbb{Z}} \delta\left(a(t - \frac{n}{a})\right)$$

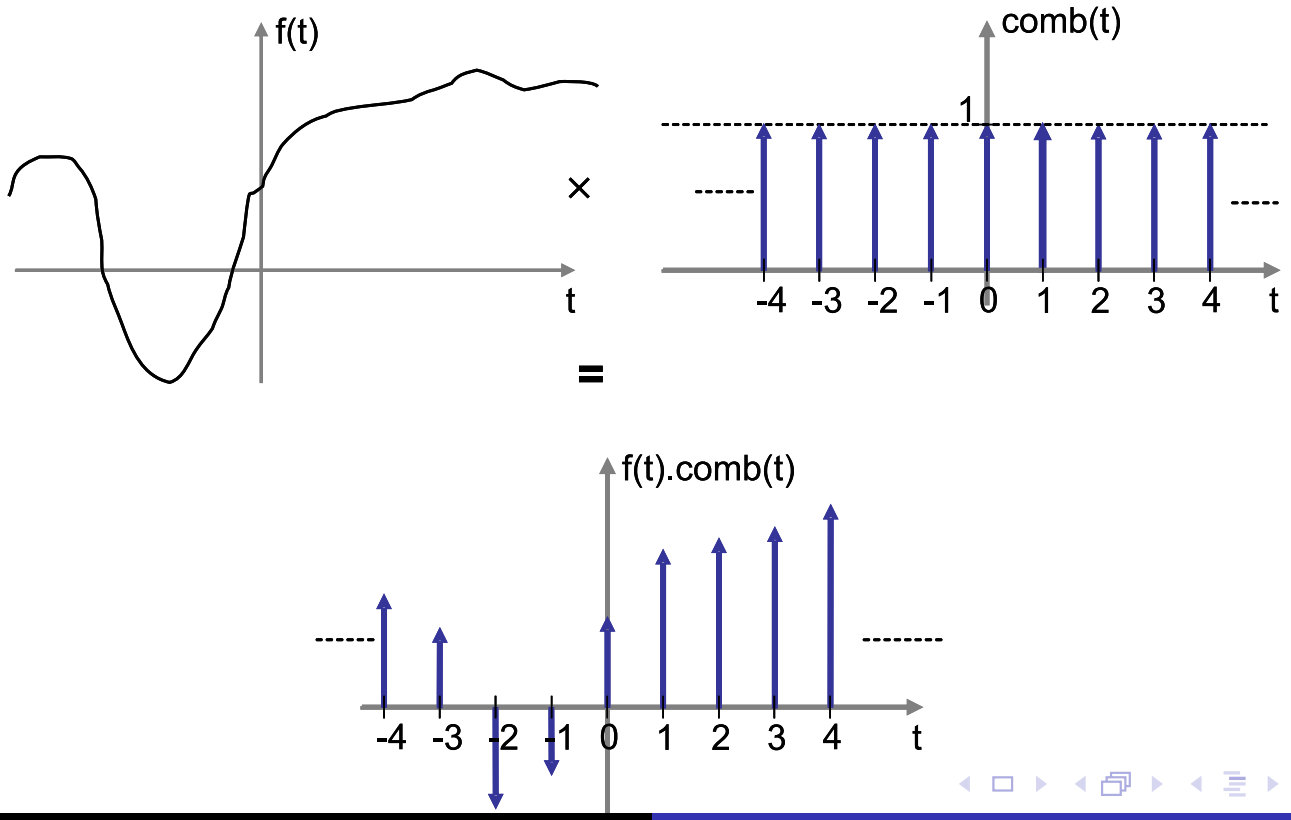
$$= \sum_{n\in\mathbb{Z}} \frac{1}{|a|}\delta\left(t - \frac{n}{a}\right) = \frac{1}{|a|}\sum_{n\in\mathbb{Z}} \delta\left(t - \frac{n}{a}\right)$$



- The time separation between two consecutive impulses is now $1/a$ rather than 1 for the comb function.

The Comb function (5)

Multiplication by the comb function:

$$f(t)comb(t) = f(t)\sum_{n\in\mathbb{Z}}\delta(t-n) = \sum_{n\in\mathbb{Z}}\underbrace{f(t)\delta(t-n)}_{=f(n)\delta(t-n)}$$

$$= \sum_{n\in\mathbb{Z}} f(n)\delta(t-n)$$